Simulating Mechanical Energy Transfer in Rowing

A Simplified One-Dimensional Model

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Abstract

This paper uses a simple model to examine how periodic thrust and quadratic drag affect a rowing shell's speed. A prescribed square-wave force accelerates a single mass during the drive and vanishes during recovery. The results illustrate speed fluctuations and energy accounting during a 20-second startup. The calculation does not measure rower efficiency or validate performance against experimental data.

1 Background and scope

Rowing involves moving body segments, oar-water interactions and hull motion that cannot be resolved by this single-mass model. A biomechanical review by Baudouin and Hawkins discusses performance factors and velocity fluctuations [1]; it provides context rather than validation of the present parameter choices. A useful next step is to fit time-resolved thrust and speed to independent measurements, assess parameter uncertainty, and compare predictions with held-out strokes. A longer simulation with a cycle-to-cycle convergence criterion would distinguish startup behavior from periodic operation. Comparisons of stroke rates should specify whether mean thrust, work per stroke, or average power is held fixed.

2 Model

The boat and crew are represented by a single mass m with speed v. The equation of motion is

$$m\frac{dv}{dt} = F(t) - kv^2. \tag{1}$$

For the baseline, $m = 766.5$ kg, $k = 40$ kg/m, and the prescribed thrust is 3000 N for 0.9 seconds in each 2-second cycle, followed by zero thrust. The boat starts from rest. These values are illustrative model inputs, not measurements from a crew.

The model calculates thrust work $W = \int Fv\,dt$, drag work $D = \int kv^3\,dt$ and the change in kinetic energy $\Delta K = m(v_f^2 - v_0^2)/2$. In the continuous model $W - D = \Delta K$. Thrust work on the modeled mass is not the rowers' total mechanical or metabolic work. The ratio $\Delta K/W$ describes how much input work remains as kinetic energy at the end of this startup interval; it is not rowing efficiency and depends on the interval chosen.

3 Method and results

The equation is integrated with explicit Euler steps of 0.01 seconds for 20 seconds. The baseline average speed over the latter half is 5.785 m/s and the maximum is 6.759 m/s. Thrust work is approximately 141.2 kJ, drag work 132.3 kJ and kinetic energy gain 9.17 kJ. The gain is 6.49% of thrust work. The difference of about -316 J in $W - D - \Delta K$ is a numerical residual from the discrete scheme.

At a 0.005-second step the residual is about -158 J; at 0.001 seconds it is about -31 J. The corresponding latter-half average speeds are 5.782 and 5.781 m/s. This convergence check addresses discretization, not the physical accuracy of the inputs.

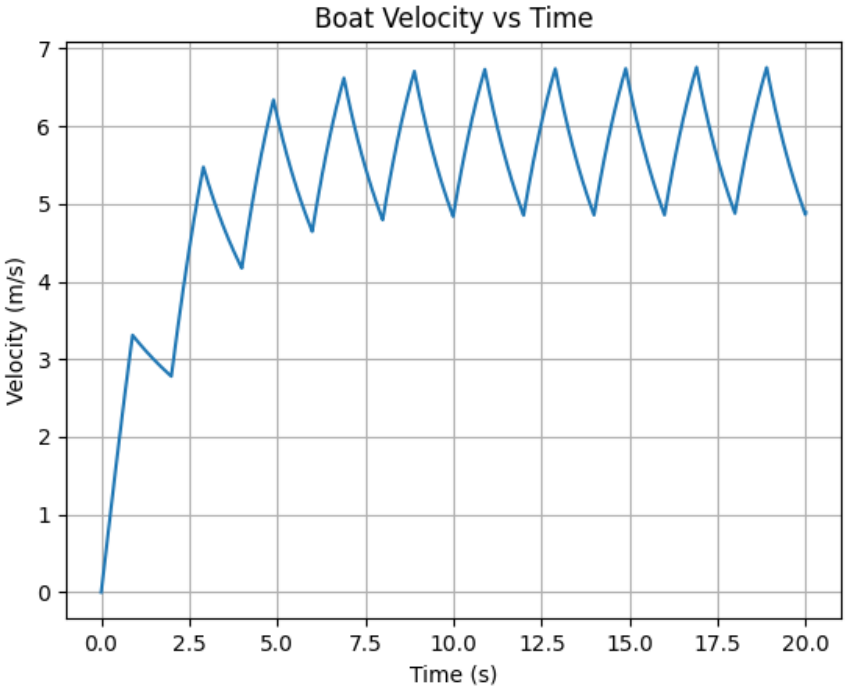


Figure 1: Baseline speed during the 20-second simulation.

4 Parameter comparisons and limitations

With the default step, halving drag raises the latter-half average speed to 8.162 m/s; halving mass gives 5.719 m/s over the same finite interval. Shortening the cycle from 2.0 to 1.5 seconds gives 6.673 m/s, but keeping the drive at 0.9 seconds also increases the drive fraction and input. This is therefore not an isolated comparison at constant work or power.

The model excludes blade slip, rower motion relative to the shell, varying force profiles, wind and waves. It has not been calibrated or compared with measured trajectories. More realistic thrust and drag measurements, timestep checks and experimental validation would be needed before applying the numerical predictions to a crew.

5 Stroke cycle and energy accounting

With cycle period $T = 2$ s and drive duration $d = 0.9$ s, the prescribed force is F_d when $0 \leq t \bmod T < d$, and zero otherwise. The resulting duty fraction is $d/T = 0.45$. The force acts on the combined modeled mass; it is not a direct measurement of force at an oar handle or a physiological power estimate.

Multiplying the equation of motion by v and integrating gives the continuous energy balance $W - D = \Delta K$. The discrete calculation evaluates force and power at the beginning of each step; the squared velocity update introduces a step-dependent residual. A small residual is a necessary consistency check, but does not verify the drag coefficient or thrust model. Velocity is clipped at zero in the code, so an energy balance would also need to account for that operation if it became active. It is not active in the reported baseline.

6 Sensitivity profiles

Figures 2–4 show the complete transient profiles rather than only the summary averages. The mass comparison changes acceleration and stroke-to-stroke fluctuations as well as the finite-interval average. The 20-second averages do not prove that a periodic steady state has been reached. Lower drag changes the relationship between thrust, speed and work, so it is not a comparison at equal mechanical power. The higher-rate case uses a longer fraction of each cycle for the drive and therefore changes more than stroke frequency.

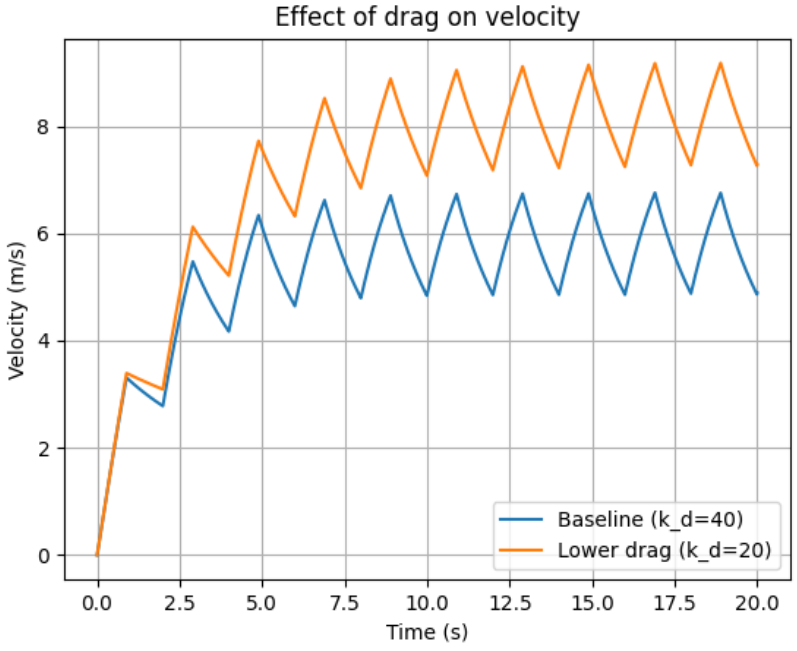


Figure 2: Velocity for baseline and lower drag.

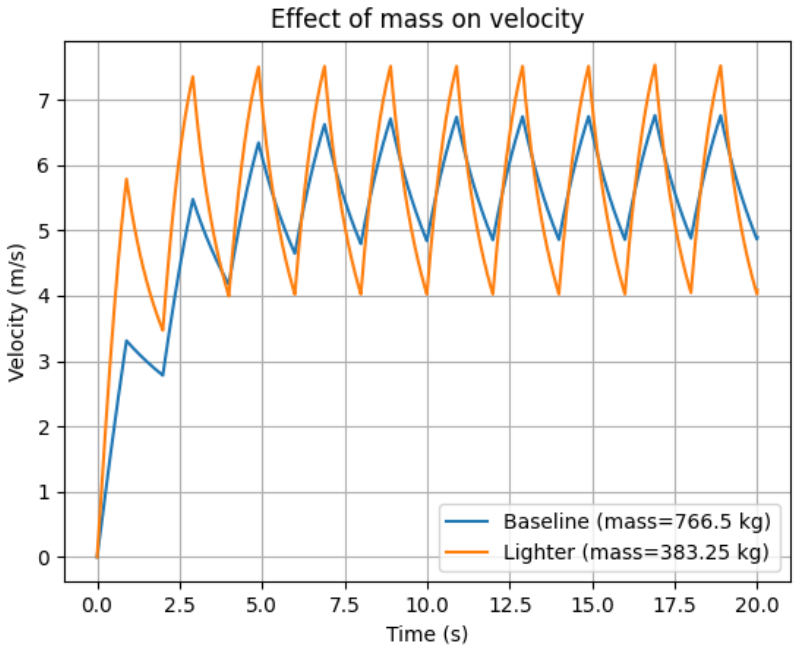


Figure 3: Velocity for baseline and lower mass.

7 Moving-mass extension

The lumped model makes the hull and crew move as one mass. To examine recovery motion, I consider a reduced model in which the crew center of mass moves relative to the hull. Let m_b be the hull-fixed mass, m_r the moving mass, $M = m_b + m_r$, X_b the hull position, and x_r the crew position relative to the hull. The combined center of mass is

$$X_c = X_b + \frac{m_r}{M} x_r. \tag{2}$$

If external surge forces are represented by thrust F and drag D, differentiating twice gives

$$M\ddot{X}_b = F - D - m_r \ddot{x}_r. \tag{3}$$

A prescribed internal acceleration changes hull acceleration even though it does not change total momentum in an isolated system. The sign depends on the coordinate direction. The relation follows from the two-mass assumptions; it does not determine crew trajectory, blade force, or internal work. Measured or explicitly prescribed motion would be needed to evaluate the term. The appendix's prescribed trajectory uses a half-cosine drive followed by a cosine, fifth-order polynomial, or blended recovery. I reconstructed this component as `src/slide_kinematics.py`; tests compare its analytic derivatives with finite differences away from transitions. Position and velocity are continuous at the drive–recovery boundary, but acceleration may jump there. This component alone does not reproduce the original multibody or IMU figures.

8 Pitch and disturbance resistance: a proposed surrogate

A possible extension represents pitch θ with effective inertia, damping, and restoring stiffness:

$$I_\theta \ddot{\theta} + c_\theta \dot{\theta} + k_\theta \theta = \tau(t). \tag{4}$$

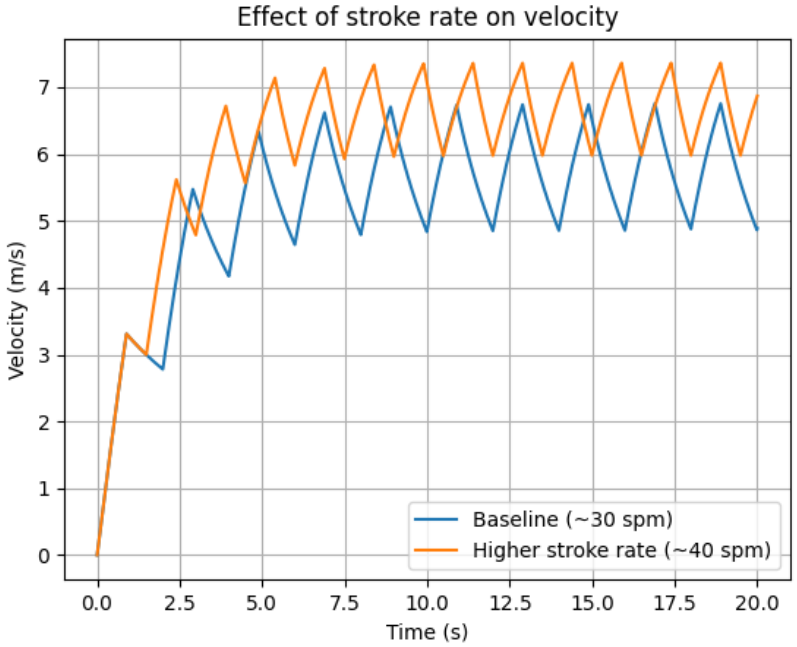


Figure 4: Velocity for two cycle periods with fixed drive duration.

A phenomenological drag law can then be written $D = kv^2(1 + \alpha\theta^2 + \beta\dot{\theta}^2)$ for forward motion. The coefficients must carry units consistent with the dimensionless multiplier and must be fitted or independently supported. The incremental work $\int kv^3(\alpha\theta^2 + \beta\dot{\theta}^2)dt$ compares terms at the same instantaneous speed. It is not automatically the difference between two coupled simulations, because changing drag can also change speed and pitch. These equations are implemented in a separate reconstructed research prototype using the parameter values in the appendix. The reconstruction passes unforced momentum and numerical refinement checks. It has not reproduced the original synthetic figures or been experimentally validated; its prescribed kinematics and pitch law are hypotheses.

9 Speed fluctuations and power

For $v = \bar{v} + u$, with $\langle u \rangle = 0$, the exact identity is

$$\langle v^3 \rangle = \bar{v}^3 + 3\bar{v}\langle u^2 \rangle + \langle u^3 \rangle. \tag{5}$$

For nonnegative speeds, convexity implies $\langle v^3 \rangle \geq \bar{v}^3$. Under this model a variable speed at fixed mean speed and fixed drag coefficient requires at least as much average drag power as constant speed. This comparison does not establish that reducing hull speed fluctuations always improves race performance: changing technique may alter propulsion, crew motion, and other losses. The model makes the energetic assumption explicit rather than providing a coaching prescription.

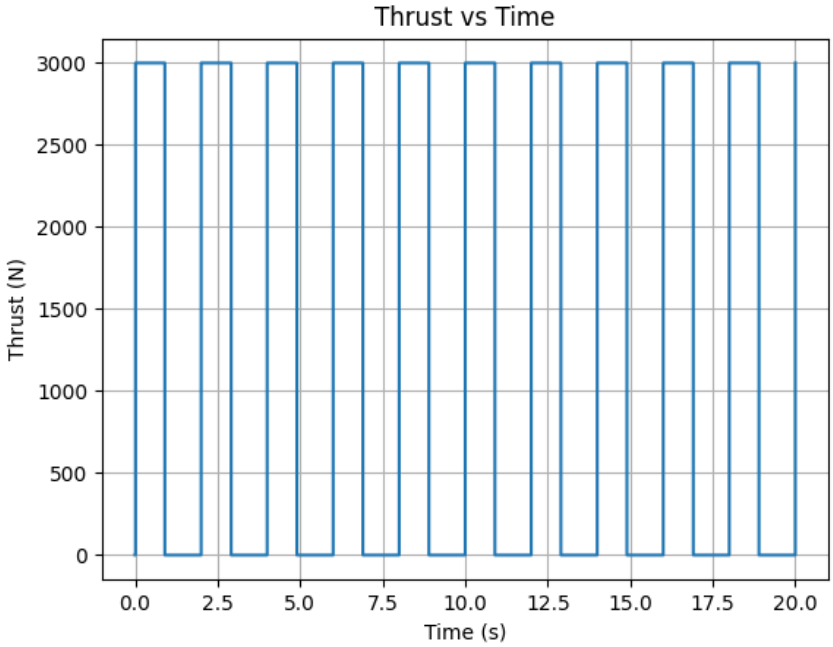


Figure 5: Prescribed thrust profile.

10 Measurement and system identification

To connect the model to measurements, a future protocol could synchronize a hull-mounted inertial measurement unit, independent speed measurements, stroke events, and a force or power proxy. Sensor orientation, gravity removal, mounting, calibration, sampling and filtering should be documented before calculating acceleration or pitch features. Integrating acceleration to obtain speed requires independent drift correction. A rowing sensor review discusses measurement choices and their constraints [2]. Events detected from a hull signal should be checked against an independent stroke reference rather than assumed to coincide with the catch or finish.

Let y_i denote measured waveforms or features and $\hat{y}_i(\psi)$ their model counterparts. A candidate fit minimizes $\sum_i (y_i - \hat{y}_i)^T \Sigma_i^{-1} (y_i - \hat{y}_i) + \lambda \Omega(\psi)$ over physically admissible parameters. The covariance or weighting matrix Σ_i and regularizer Ω must be specified before fitting. Sensitivity rank, profile likelihood or comparable diagnostics are needed to examine whether parameters can be separated; a good waveform fit alone does not prove identifiability. Validation should use held-out sessions or trials, report uncertainty and compare predictions with independently measured outcomes.

10.1 Synthetic reconstruction and checks

The source manuscript's synthetic demonstration is distinct from such empirical validation. I reconstructed uniform resampling and a small-angle complementary filter from its listing, with tests for stationary tilt and interpolation. The original listing's synthetic accelerometer uses a negative gravity sign while its preprocessing equations assume positive vertical gravity at rest. The reconstructed synthetic sensor adopts the filter's positive vertical gravity at rest; a stationary test checks this convention. Its small-angle planar acceleration approximation is not a validated full inertial sensor model.

A reconstructed acceleration-magnitude peak detector was checked on regularly spaced synthetic pulses; its peaks are candidate events, not validated catch labels. A separate least-squares prototype fits pitch-drag and trim coefficients to modeled speed and pitch-rate observations. For a four-second noiseless simulation generated by the same model, it recovered the inputs 2.5 and 18000 to the printed precision. This inverse-crime test checks implementation only; it does not establish parameter identifiability under noise, model mismatch, or real measurement conditions. In one fixed-seed synthetic trial with speed noise standard deviation 0.01 m/s and pitch-rate noise 0.002 rad/s, the fitted pitch-drag coefficient was 2.31 for a true value of 2.5. With the same data and the drag coefficient fixed 10 percent too high, the fitted coefficient fell near zero, while the optimizer still reported success. This single example illustrates model dependence, not a statistical uncertainty interval.

The synthetic trajectory in Figure 6 has large pitch rates and even brief backward hull motion from rest; it is illustrative, not a realistic prediction. The source manuscript's original synthetic figures have not been independently reproduced. An adapted recovery-blend sweep follows the listing's three plotted metrics rather than the cost-per-distance quantity described in the original caption. In this reconstruction, increasing the blend from 0 to 1 increases the computed jerk RMS, pitch RMS, and speed coefficient of variation. These outcomes depend on the prescribed trajectory, model parameters, and numerical timestep; the acceleration discontinuities also make numerical jerk sensitive to sampling. They do not establish a technique recommendation.

The original cohort example ranks recordings by the mean of an index standardized separately within each recording. That mean is zero to numerical precision, so the resulting ordering does not establish differences between athletes. A separate archived beta-force implementation normalizes each input array by its own maximum; called on one time point at a time, it becomes constant at nonzero phases and undefined at zero. I therefore do not use that version for the simulations presented here.

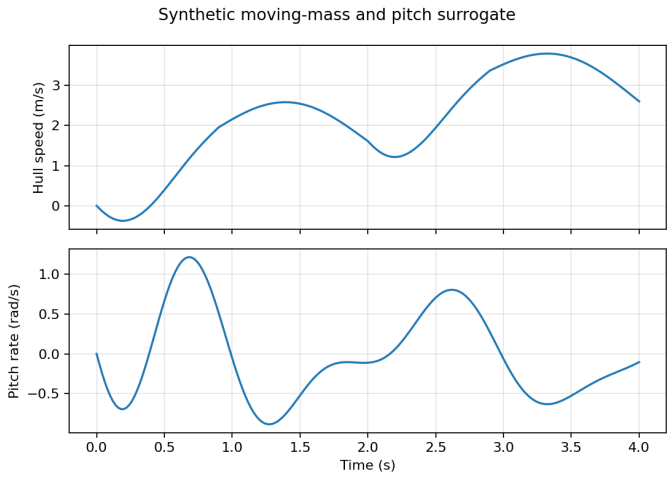


Figure 6: Synthetic trajectory from the reconstructed moving-mass and pitch surrogate, using illustrative manuscript parameters.

11 Conclusion

The baseline simulation illustrates how prescribed thrust and drag shape the speed and energy balance of a rowing shell during startup. The moving-mass, pitch, and sensor models extend that framework, but their parameters and predictions require independent measurement. The numerical and synthetic checks support the implementation under the stated assumptions; they do not demonstrate a real-world efficiency gain. Future work should obtain synchronized force, speed, and inertial data, test sensitivity and identifiability, and evaluate predictions on held-out sessions.

References

[1] A. Baudouin and D. Hawkins, "A biomechanical review of factors affecting rowing performance," *British Journal of Sports Medicine*, 36, 396–402, 2002. `https://bionics.seas.ucla.edu/education/Rowing/Math_Model_2002_01.pdf`.

[2] M. T. O. Worsey, H. G. Espinosa, J. B. Shepherd and D. V. Thiel, "A Systematic Review of Performance Analysis in Rowing Using Inertial Sensors," *Electronics*, 8(11), 1304, 2019. `https://doi.org/10.3390/electronics8111304`.